G11 Functions — Transformations & Inverse Functions Practice

Name: ______________________ Date: ______________

Graphing Reminder (Five-point plotting): For each graphing question, use the **five-point method**. Plot and label **at least five key points** (e.g., vertex, intercepts, and other critical points). Choose points that keep calculations clean.

Part A: Transformations (3 questions)

1) Square Root Transformation (Graphing)

Let $f(x) = \sqrt{x}$. Define

$$g(x) = -3\,f\left(\tfrac{1}{2}(x+4)\right) + 5.$$

- (a) List **all transformations** from $f(x)$ to $g(x)$ in the correct order (include direction and factor).
- (b) On the Cartesian plane provided, graph $g(x)$ using the **five-point method**. Label at least **five points**.
- (c) State the **domain** and **range** of $g(x)$.

Transformations (notes/work):

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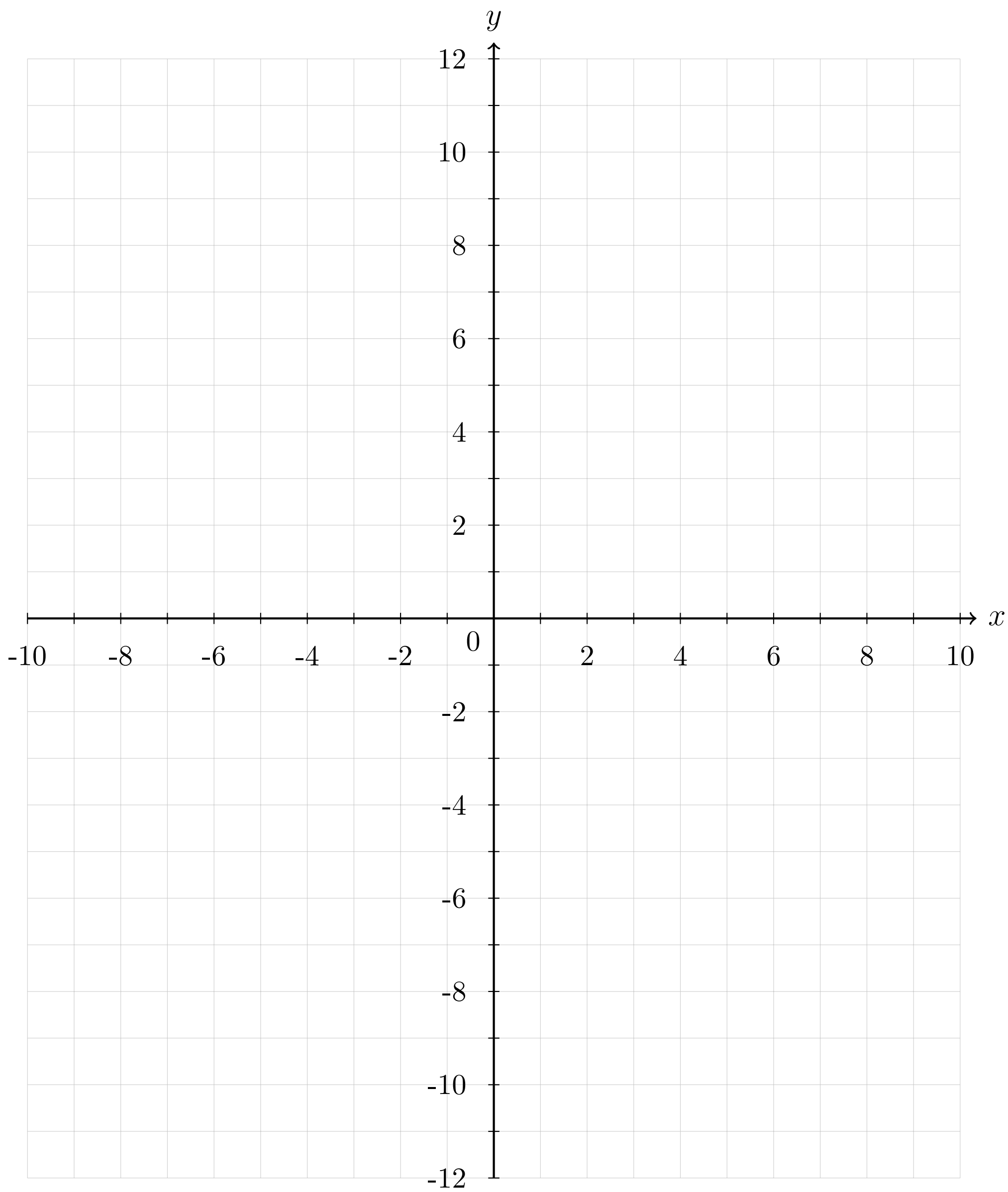
Graph for Question 1: *(Use five-point plotting)*



Domain: ______________________________ Range: ______________________________

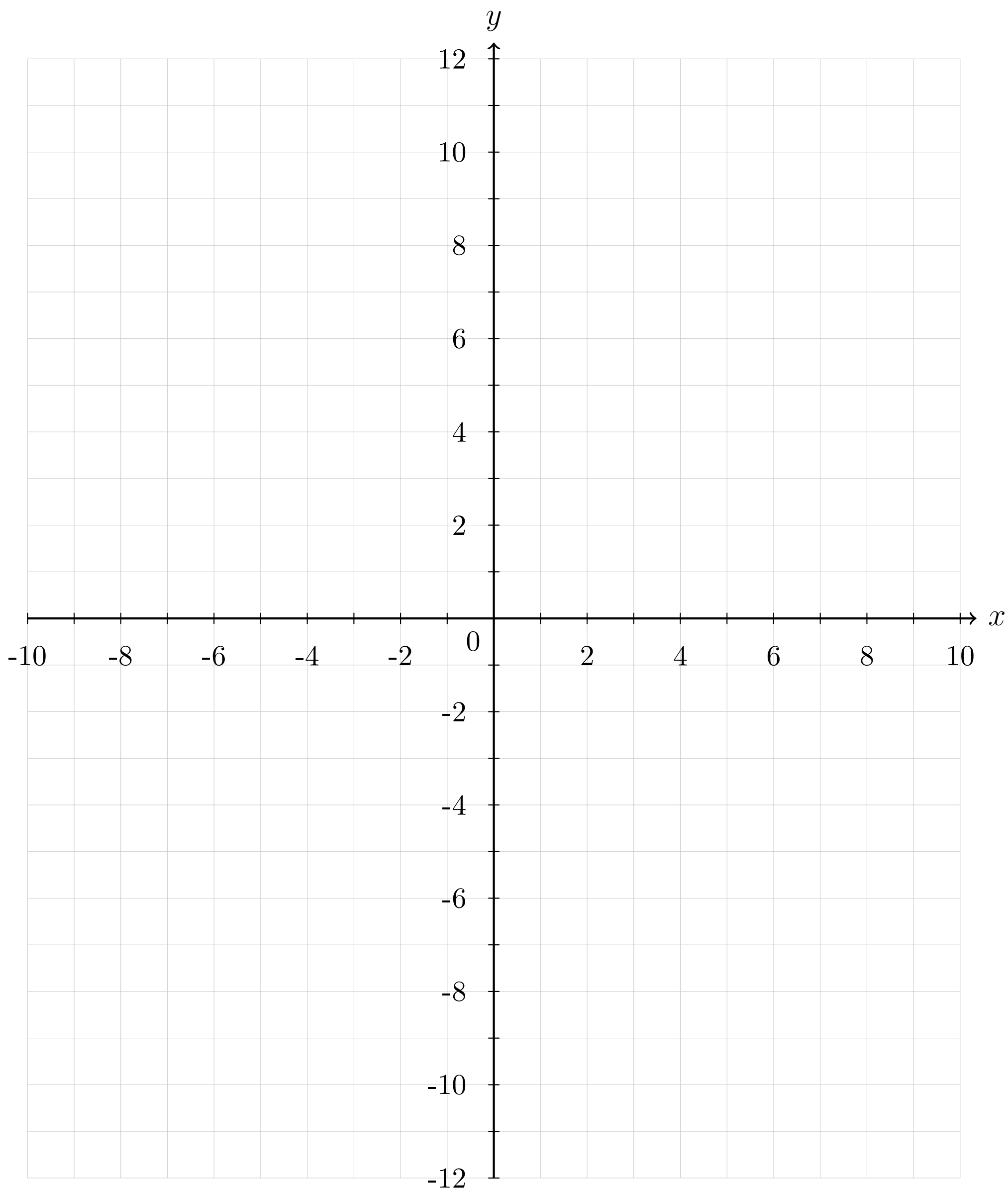
2) Absolute Value with Reflection (Graphing + Intercepts)
Let $f(x) = |x|$. Define

$$h(x) = 2\,f(-3(x-2)) - 7.$$

- (a) List **all transformations** from $f(x)$ to $h(x)$ in the correct order.
- (b) On the Cartesian plane provided, graph $h(x)$ using the **five-point method**. Label the **vertex** and at least **four additional points**.
- (c) Find the x**-intercepts** of $h(x)$ (show algebra clearly).
- (d) State the **domain** and **range** of $h(x)$.

Transformations (notes/work):

Graph for Question 2: *(Use five-point plotting)*



x-intercepts (show work):

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Domain: ____________________ Range: ____________________

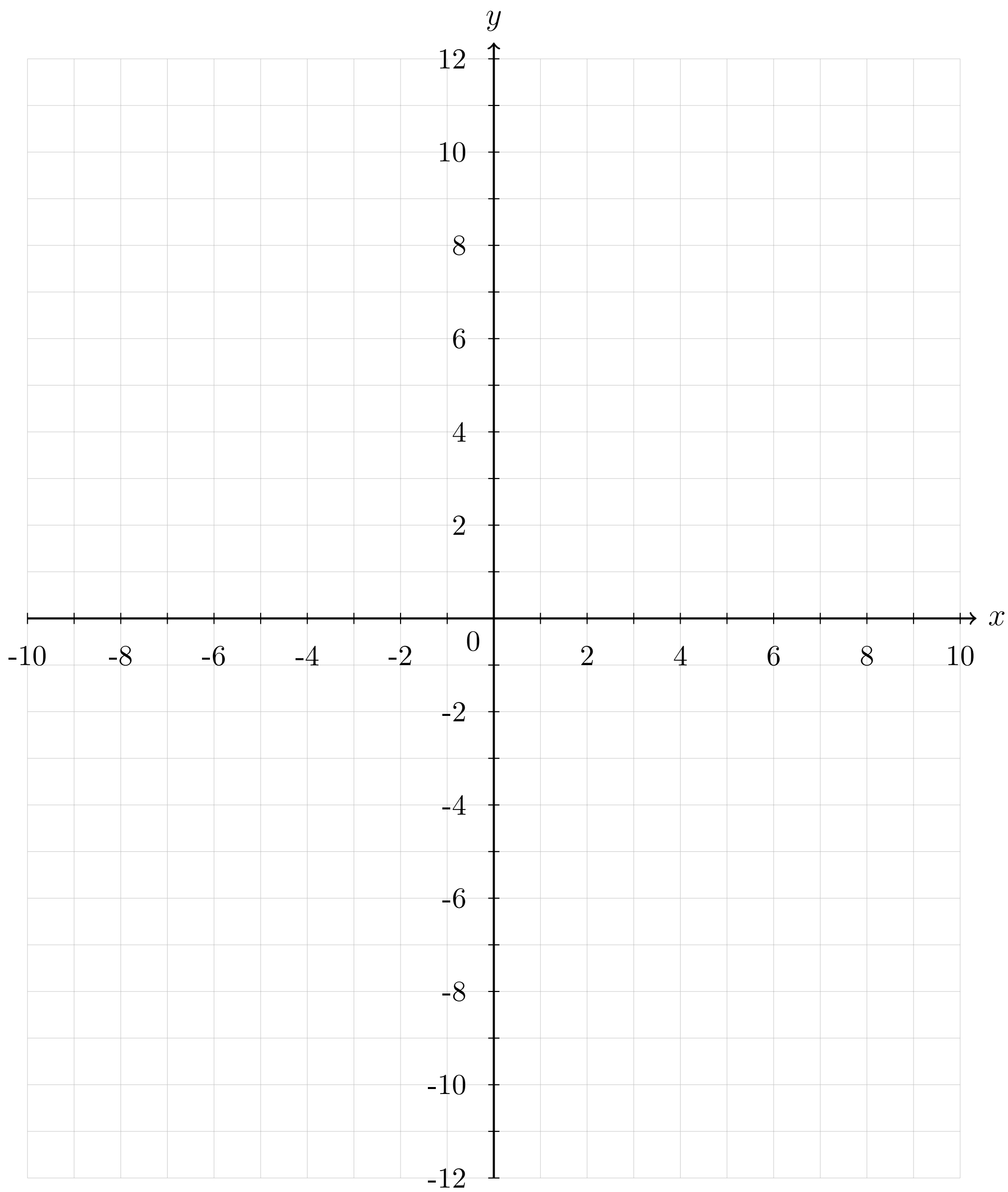
3) Determine a Transformation from Conditions (Graphing)

A transformed parabola $q(x)$ comes from the parent function $f(x) = x^2$ and satisfies:

- it opens **downward**,
- its vertex is $(3, -2)$,
- it passes through $(5, -10)$.
- (a) Write the model $q(x) = a(x-3)^2 - 2$.
- (b) Use the point $(5, -10)$ to solve for a (show all steps).
- (c) Write the final equation for $q(x)$.
- (d) Graph $q(x)$ on the Cartesian plane using the **five-point method**.
- (e) State the **range** of $q(x)$.

Work (solve for a and finalize $q(x)$):

Graph for Question 3: *(Use five-point plotting)*



Range: ______________________________

Part B: Inverse Functions (5 questions)

General instructions: To find an inverse, show the standard steps:

$$y = f(x) \ \rightarrow \ x = f(y) \ \rightarrow \ \text{solve for } y \ \rightarrow \ f^{-1}(x).$$

Always state domain/range when asked, and **verify** when required.

4) Linear Inverse + Verification

$$f(x) = -\frac{3}{5}x + \frac{7}{2}$$

- (a) Find $f^{-1}(x)$ (show all steps).
- (b) Verify $f(f^{-1}(x)) = x$ by substitution and simplifying.
- (c) State the domain and range of f and f^{-1}.

Work:

5) Radical Inverse + Domain/Range

$$g(x) = \sqrt{2x - 5} - 3$$

- (a) Find the domain and range of $g(x)$ (use inequalities).
- (b) Find $g^{-1}(x)$ (show all steps, including squaring).
- (c) State the domain and range of $g^{-1}(x)$.

Work:

6) Quadratic Inverse with Domain Restriction

$$h(x) = (x - 4)^2 + 1$$

- (a) Explain briefly why $h(x)$ is not one-to-one (horizontal line test).
- (b) Choose and state a domain restriction (e.g., $x \geq 4$ *or* $x \leq 4$).
- (c) Find $h^{-1}(x)$ under your restriction (show steps).
- (d) State the domain and range of $h^{-1}(x)$.

Work:

7) Rational Function Inverse (Algebra-heavy)

$$m(x) = \frac{2x - 3}{x + 5}$$

- (a) State the domain of $m(x)$.
- (b) Find $m^{-1}(x)$ (show all algebra steps carefully).
- (c) State the domain of $m^{-1}(x)$.
- (d) Verify using a numerical check: compute $m(1)$ and then $m^{-1}(m(1))$.

Work:

8) Transformation + Inverse (Full Process)

$$t(x) = 2(x - 1)^2 - 8$$

- (a) State the vertex and range of $t(x)$.
- (b) Choose a domain restriction so that t is one-to-one (state it clearly).
- (c) Find $t^{-1}(x)$ under your restriction (show steps).
- (d) State the domain and range of $t^{-1}(x)$ and compare with your answers in (a).

Work:

End of worksheet. Make sure every inverse is simplified and every graph has at least five labeled points.